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POWER BI MAJOR CAPSTONE PROJECT  
**END TO END**  
**POWER BI PROJECT GUIDE**

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| <b>PROJECT TITLE</b> | Student Social Media & Mental Health Analytics |
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| <b>PROJECT CATEGORY</b> | Education Analytics<br>Mental Health Analytics<br>Behavior Analytics |
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| <b>INSTITUTE NAME</b> | Infinity Code Nexus |
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| <b>TRAINER / GUIDE</b> | Mr.Anamaina Anand Kumar |
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| <b>CREATED BY</b> | Nagavelli Manikanta<br>Chary |
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| <b>TOOLS USED</b> | Power BI Desktop Power<br>Query, DAX & Power BI<br>Service |
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| <b>SUBMISSION DATE</b> | [13/06/2026] |
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Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

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## Project Overview

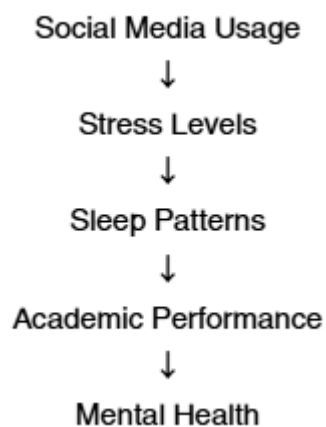
### Project objectives :

Develop an interactive dashboard to analyze the impact of social media usage on student mental health.

Participants are expected to:

- ✓ Analyze student behavior
- ✓ Measure social media usage
- ✓ Evaluate stress levels
- ✓ Understand mental health patterns
- ✓ Generate actionable insights
- ✓ Present recommendations

Convert raw student data into meaningful insights that help understand the relationship between.



## Business Problem

### Educational institutions and student counselors want to understand:

1. How social media affects mental well-being.
2. How screen time impacts academic performance.
3. How stress levels vary across different student groups.
4. How sleep patterns influence mental health.
5. Which social media platforms have the highest usage.

### Participants must answer:

1. What is the average mental health score?
2. What is the average social media usage per day?
3. Which platform is most used?
4. Which student groups report higher stress levels?
5. How does sleep affect mental health?
6. How does study time compare with social media usage?
7. Which countries show higher mental health scores?

## Dataset Information

**Dataset Name** : Student Social Media and Mental Health Impact

| Item                     | DATASET DETAILS     | Records |
|--------------------------|---------------------|---------|
| <input type="checkbox"/> | Approximate Records | 5,000+  |



## POWER QUERY TRANSFORMATIONS

### DATA CLEANING ACTIVITIES

- ✓ Removed Duplicate Records
- ✓ Handled Missing Values
- ✓ Corrected Data Types

### DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns
- ✓ Created Custom Columns
- ✓ Replaced Values
- ✓ Split Columns
- ✓ Merged Columns
- ✓ Removed Unnecessary Columns

### CUSTOM COLUMNS CREATED

- ✓ Age Group
  - ✓ Usage Level
  - ✓ Study Level Category
  - ✓ Sleep Category
  - ✓ Stress Category
  - ✓ Mental Health Status
  - ✓ Platform Usage Type
  - ✓ Lifestyle Balance Index
-

# DATA MODELING

## Star Schema

### Purpose:

FACT TABLE

student\_Fact

### DIMENSION TABLES

Dim\_Location

Dim\_Tech\_Field

Dim\_Job\_Title

Dim\_Education

Dim\_Work\_Arrangement

### RELATIONSHIPS

One-to-Many Relationships

### FILTER DIRECTION

Single Direction

### MODEL BENEFITS

Improved Performance

Easy DAX Development

Better Scalability

## DAX MEASURES USED

### Total Students

Total Students =

COUNTROWS('Student Social Media And Mental Health Impact')

### PURPOSE

calculates the total number of student records

**AVERAGE USAGE HOURS**

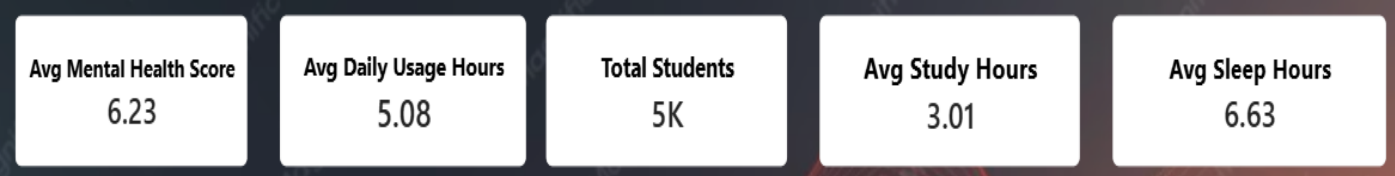
Avg Daily Usage Hours =  
AVERAGE('Student Social Media And Mental Health Impact'[Avg\_Daily\_Usage\_Hours])

**PURPOSE**

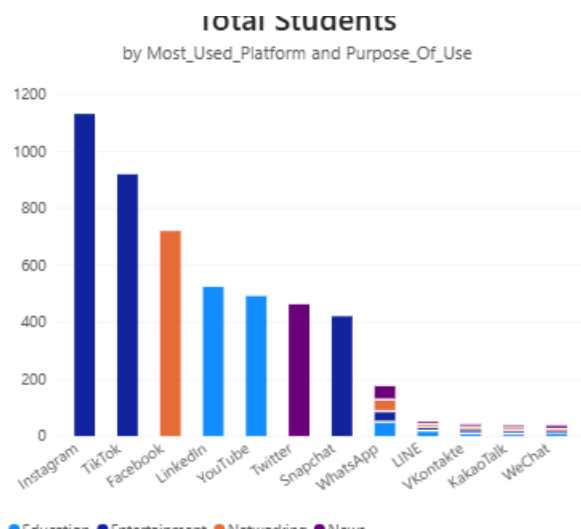
Average number of hours students spend daily on social media.

**VISUALIZATIONS USED**

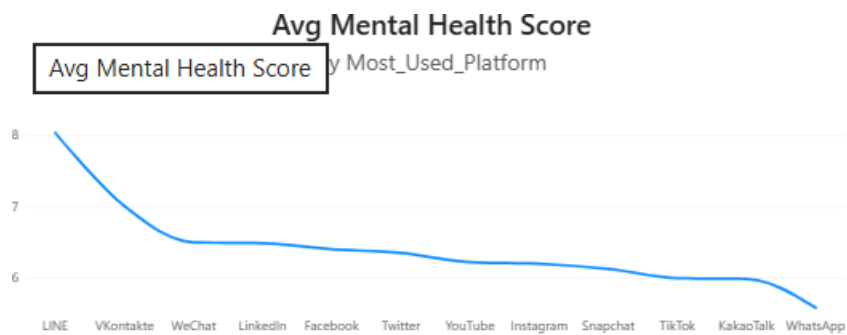
**KPI CARDS:**



**BAR CHART :**



## LINE CHART:



## DASHBOARD SCREENSHOTS

## EXECUTIVE OVERVIEW:

# Executive Overview

Country

All

Gender

All

Most\_Used\_Platform

All

Avg Mental Health Score  
6.23

Avg Daily Usage Hours  
5.08

Total Students  
5K

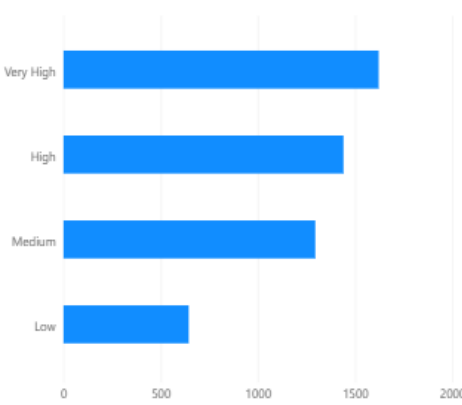
Avg Study Hours  
3.01

Avg Sleep Hours  
6.63

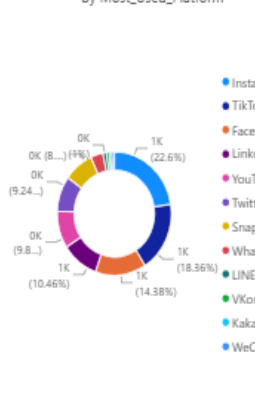
Avg Mental Health Score  
by Country



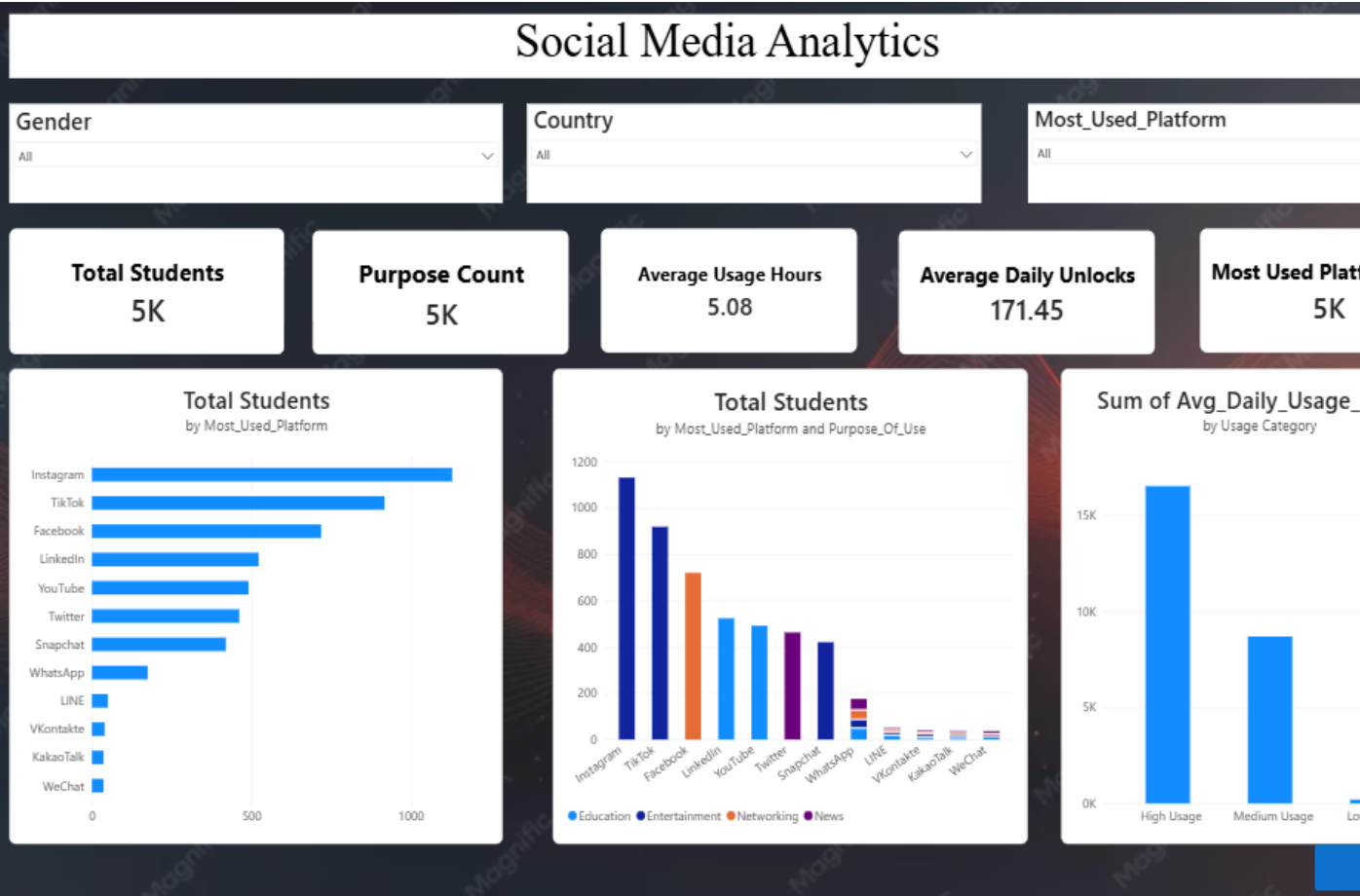
Total Students  
by Stress\_Level



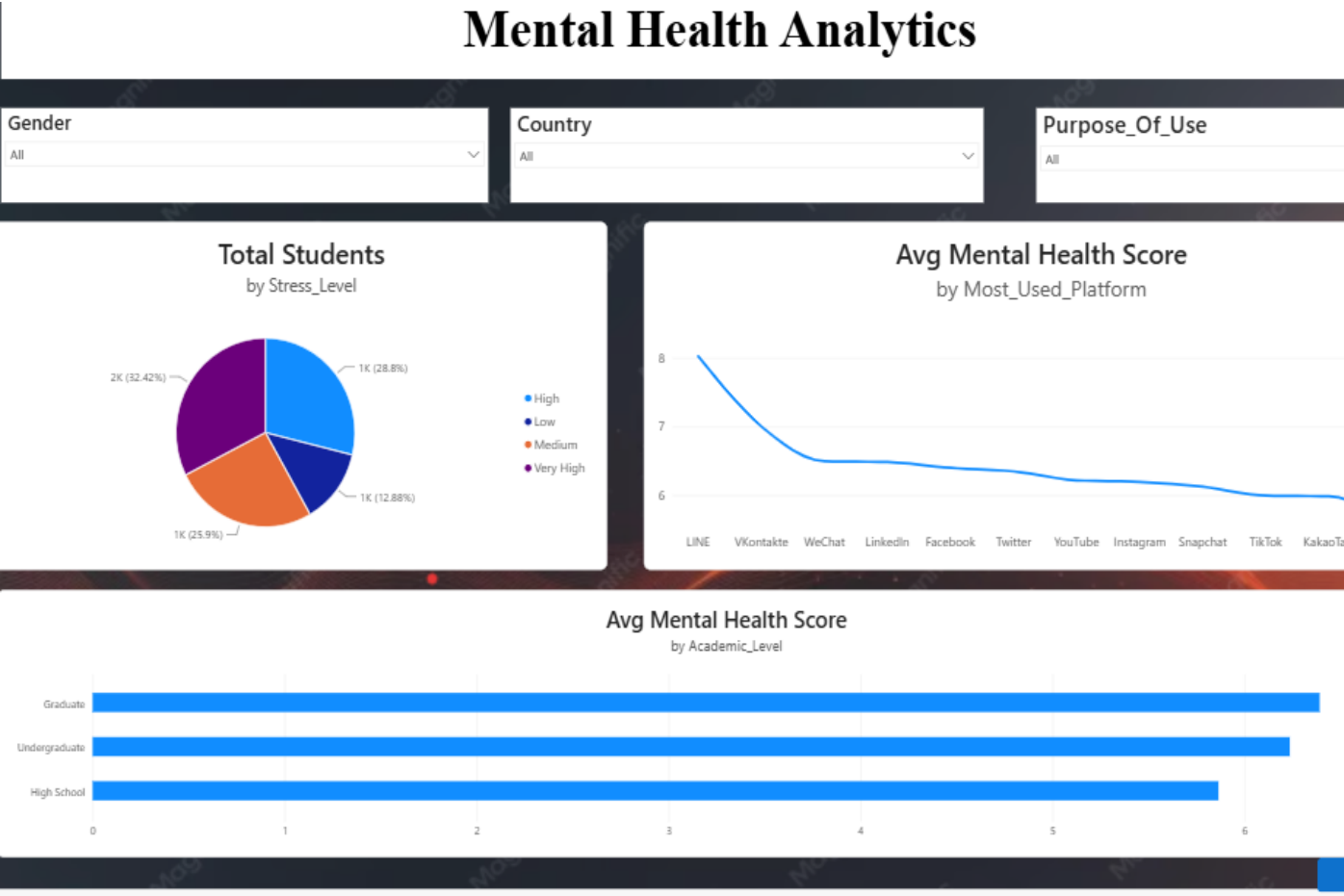
Total Students  
by Most\_Used\_Platform



SOCIAL MEDIA ANALYTICS:



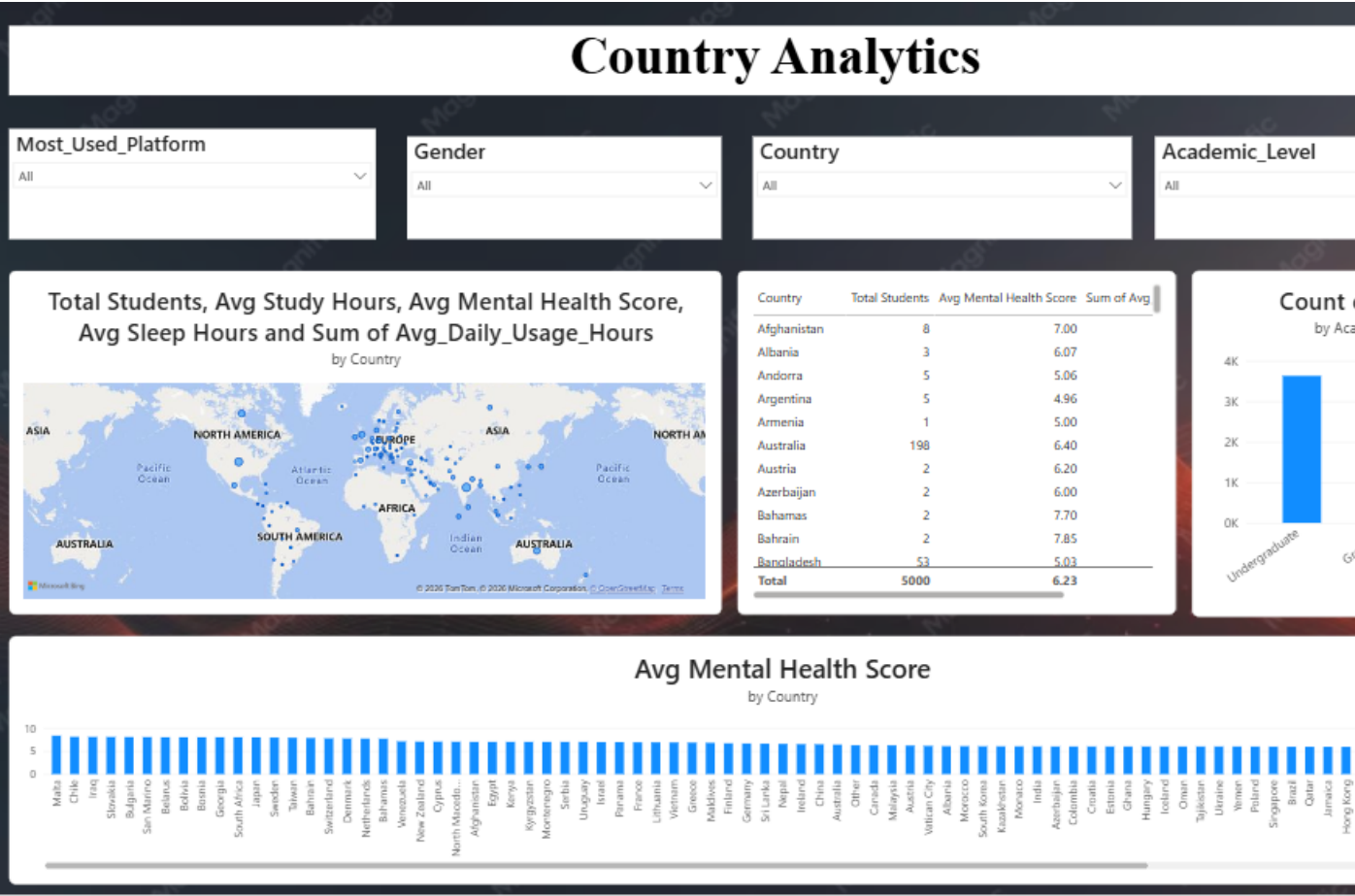
MENTAL HEALTH ANALYTICS:



LIFESTYLE ANALYTICS:



COUNTRY ANALYTICS:



KEY INSIGHTS

INSIGHT-1:

Reveals how much time students spend daily on social media and identifies low, moderate, or high usage patterns.

### **INSIGHT-2:**

Demographic Insight: Shows differences in average usage hours across gender, country, and academic level.

### **INSIGHT-3:**

Mental Health Insight: Helps understand the relationship between higher social media usage, increased stress levels, and changes in mental health scores.

## RECOMMENDATIONS

### **RECOMMENDATION-1**

**Promote Healthy Usage Habits:** Encourage students to limit daily social media time and maintain balanced screen-time routines.

### **RECOMMENDATION-2**

**Target High-Usage Groups:** Design awareness and support programs for demographic or academic groups with higher average usage hours.

### **RECOMMENDATION-3**

**Integrate Mental Health Support:** Combine usage insights with stress and mental health data to provide early interventions and counseling support.

## FEATURES IMPLEMENTED

**Purpose :** *Participants must implement the following concepts.*

### **DRILL DOWN :**

*Examples :*

*Country*



*Gender*



*Academic Level*

## DRILL THROUGH :

### Examples

*Country*



*Detailed Student Analysis*

*Platform*



*Platform Detail Analysis*

## CONDITIONAL FORMATTING :

Apply color logic using DAX.

Green = Healthy Score

Orange = Average Score

Red = Low Score

## SLICERS :

**Required Slicers :** Country , Gender , Academic Level and Most Used Platform.

## Storytelling

### SCENARIO:

With the rapid rise of social media usage among students, understanding daily screen-time behavior has become critical. This analysis focuses on measuring **average daily usage hours** to establish a clear baseline of student engagement. By examining usage across **gender, country, and academic level**, the report uncovers patterns that highlight high-usage student groups. The data further connects screen time with **stress levels and mental health scores**,

revealing potential risks of excessive usage. Interactive visuals enable deeper exploration of these trends, helping stakeholders identify problem areas. Overall, the story emphasizes the importance of **balanced digital habits** for student well-being.

#### **PROBLEM:**

1. Lack of clarity on students' average daily social media usage.
2. Difficulty identifying high-usage student groups.
3. Limited understanding of usage differences across demographics.
4. Unclear relationship between screen time and mental health.
5. Absence of data-driven insights to support wellness initiatives.

#### **ANALYSIS APPROACH:**

1. Calculate average daily usage hours using DAX.
2. Use card visuals to show overall usage KPIs.
3. Apply bar charts to compare usage across demographics.
4. Analyze relationships between usage, stress, and mental health.
5. Enable slicers and filters for interactive, detailed exploration.

#### **KEY FINDINGS:**

Students show moderate to high average daily social media usage, indicating strong digital engagement.

Usage levels vary noticeably by gender, country, and academic level.

Certain student groups display consistently higher screen time, identifying potential risk segments.

Higher daily usage hours often align with increased stress and mental health concerns.

#### **IMPACT:**

Higher average daily social media usage highlights increased risk of stress and reduced mental well-being among students.

The analysis helps identify high-usage student groups for targeted support and interventions.

Insights support data-driven decisions for digital wellness and mental health programs.

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## **PROJECT DEPLOYMENT**

## **DEPLOYMENT OBJECTIVE:**

Deploy the Power BI solution to enable stakeholders to access interactive dashboards and business insights in real time.

## **DEPLOYMENT PLATFORM**

Power BI Service

### **1. DEPLOYMENT PROCESS:**

Power BI Service

#### **STEP-1**

Validate Data Model and DAX Measures in Power BI Desktop.

#### **STEP-2**

Optimize Dashboard Layout, Performance, and User Experience.

#### **STEP-3**

Publish PBIX File to Power BI Service.

#### **STEP-4**

Configure Dataset Refresh Settings.

#### **STEP-5**

Create Dashboard from Published Reports.

#### **STEP-6**

Apply Security and Access Permissions.

#### **STEP-7**

Share Dashboard with Business Stakeholders

## **BENEFITS :**

Centralized Reporting Platform.

Interactive Workforce Analytics.

Real-Time Performance Monitoring.

Improved Strategic Planning

Faster Data-Driven Decisions.

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## Conclusion

Students spend considerable time on social media daily, with usage varying across demographic and academic groups.